

C03 AT2: DATA INTERPRETATION

DSA0405:FUNDAMENTALS OF DATA SCIENCE

11. Standardization / Normalization

Step 1: Why is standardization needed?

A distance-based algorithm calculates distances using numerical values. Income has a much larger numerical scale than Age, so Income can dominate the distance calculation.

For example:

Age difference = 5

$$5^2 = 25$$

Income difference = ₹50,000

$$(50,000)^2 = 2,500,000,000$$

Therefore, without standardization, Income gets much more influence than Age.

Step 2: Write the z-score formula

$$z = (x - \text{Mean}) / \text{SD}$$

Step 3: Substitute the values

Given:

$$x = 45$$

$$\text{Mean} = 40$$

$$\text{SD} = 12$$

$$z = (45 - 40) / 12$$

$$z = 5 / 12$$

$$z = 0.4167$$

$$z \approx 0.417$$

Step 4: Interpret the result

Age 45 is approximately 0.417 standard deviations above the mean.

Final Answer

Failure to standardize makes the larger-scale Income feature dominate distance calculations. Standardization puts the variables on a comparable scale.

$$Z\text{-score} = 0.417$$

12. Distribution Shape from Summary Statistics Alone

Step 1: Look at mean, median and mode

$$\text{Mean} = 45$$

$$\text{Median} = 45$$

$$\text{Mode} = 45$$

Since $\text{Mean} = \text{Median} = \text{Mode} = 45$, the distribution is centered around 45 and is approximately symmetric.

Step 2: Look at skewness

$$\text{Skewness} \approx 0$$

A skewness value near zero indicates that the distribution has no significant left or right skew.

Step 3: Look at kurtosis

$$\text{Kurtosis} = 5.8$$

For a normal distribution, Pearson kurtosis ≈ 3 .

Since:

$$5.8 > 3$$

the distribution is leptokurtic.

Step 4: Interpret the kurtosis

A leptokurtic distribution has heavier tails and a greater tendency to contain extreme values than a normal distribution.

Final Answer

The distribution is approximately symmetric because Mean = Median = Mode and skewness ≈ 0 . However, kurtosis = 5.8 shows that it is leptokurtic, with heavier tails and more extreme values than a normal distribution.

13. Grouped Data – Mean and Variance

Step 1: Find the class midpoints

Midpoint $x = (\text{Lower limit} + \text{Upper limit}) / 2$

Class Interval	Frequency (f)	Midpoint (x)
0–10	5	5
10–20	8	15
20–30	15	25
30–40	12	35
40–50	10	45

Step 2: Choose assumed mean and class width

Assumed mean, $A = 25$

Class width, $h = 10$

Step-deviation formula:

$$d' = (x - A) / h$$

Step 3: Calculate d' , fd' and fd'^2

x	f	d'	fd'	d'^2	fd'^2
5	5	-2	-10	4	20
15	8	-1	-8	1	8
25	15	0	0	0	0
35	12	1	12	1	12
45	10	2	20	4	40
Total	50		14		80

Step 4: Calculate the mean

$$\text{Mean} = A + h(\Sigma fd' / \Sigma f)$$

$$\text{Mean} = 25 + 10(14 / 50)$$

$$\text{Mean} = 25 + 10(0.28)$$

$$\text{Mean} = 25 + 2.8$$

$$\text{Mean} = 27.8$$

Step 5: Calculate the variance

$$\text{Variance} = h^2[(\Sigma fd'^2 / \Sigma f) - (\Sigma fd' / \Sigma f)^2]$$

$$\text{Variance} = 10^2[(80 / 50) - (14 / 50)^2]$$

$$\text{Variance} = 100[1.6 - (0.28)^2]$$

$$\text{Variance} = 100[1.6 - 0.0784]$$

$$\text{Variance} = 100(1.5216)$$

$$\text{Variance} = 152.16$$

Final Answer

$$\text{Mean} = 27.8$$

$$\text{Variance} = 152.16$$

14. Comparing Two Distributions Using CV

Step 1: What is CV?

Coefficient of Variation is:

$$CV = (SD / \text{Mean}) \times 100$$

It measures risk relative to the average return.

Step 2: Calculate CV for Portfolio A

Given:

$$\text{Mean} = 12\%$$

$$SD = 3\%$$

$$CV_a = (3 / 12) \times 100$$

$$CV_a = 0.25 \times 100$$

$$CV_a = 25\%$$

Step 3: Calculate CV for Portfolio B

Given:

$$\text{Mean} = 18\%$$

$$SD = 5\%$$

$$CV_\beta = (5 / 18) \times 100$$

$$CV_\beta = 0.2778 \times 100$$

$$CV_\beta \approx 27.78\%$$

Step 4: Compare

Portfolio	Mean Return	SD	CV
A	12%	3%	25%
B	18%	5%	27.78%

Absolute risk

Portfolio B has:

$$5\% > 3\%$$

So B has higher absolute variability/risk.

Relative risk

CV shows:

$$27.78\% > 25\%$$

So B also has higher risk relative to its return.

Step 5: What does CV capture that SD alone does not?

SD measures absolute variability.

CV measures variability relative to the mean return.

Therefore, CV is useful for comparing distributions with different average returns.

Final Answer

CV of A = 25%

CV of B = 27.78%

Portfolio B is riskier in absolute terms and also slightly riskier per unit of return according to CV.

15. Outlier's Effect on Variance

Step 1: Identify the change in the value

Original value = 40

Incorrect value = 400

Change = $400 - 40$

Change = 360

Step 2: Effect on the mean

Number of observations = 10

Approximate increase in mean = $360 / 10$

Approximate increase in mean = 36

Original mean ≈ 42

Approximate new mean $\approx 42 + 36$

Approximate new mean ≈ 78

Therefore, the mean increases substantially.

Step 3: Effect on the variance

Variance = $\Sigma(x - \bar{x})^2 / N$

The original value 40 is close to the mean 42.

Original deviation:

$$40 - 42 = -2$$

Squared deviation:

$$(-2)^2 = 4$$

Step 4: Calculate the deviation of the incorrect value

$$400 - 42 = 358$$

$$\text{Squared deviation} = 358^2$$

$$\text{Squared deviation} = 128,164$$

Step 5: Compare the squared deviations

$$\text{Original squared deviation} = 4$$

$$\text{Incorrect squared deviation} = 128,164$$

Therefore, the incorrect value contributes an enormously larger amount to the variance.

Step 6: Why is variance affected more than the mean?

Mean uses the values directly:

$$\text{Mean} = \Sigma x / N$$

Variance uses squared deviations:

$$\text{Variance} = \Sigma (x - \bar{x})^2 / N$$

Because the deviation is squared, an extreme value has a disproportionately large effect on variance.

Final Answer

Mean: increases substantially.

Variance: increases enormously.

Reason: variance squares deviations, so it is much more sensitive to outliers than the mean.